

MOHSEN MINAEI

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PROFESSIONAL SUMMARY

Senior Research Scientist specializing in security, privacy, and blockchain technologies, with expertise from Visa and Microsoft. Led research on CBDCs, programmable payment channels, and blockchain interoperability at Visa. At Microsoft, contributed to fraud detection and machine learning-driven user retention. These innovations resulted in 4 external pilots, 10+ patents, and 20+ publications in top security and cryptography conferences, focusing on privacy-preserving financial solutions and scalable blockchain systems.

EDUCATION

Purdue University, Indiana, USA

- Ph.D. in Computer Science, Advisor: Aniket Kate 2014 - 2020
- M.S. in Computer Science 2014 - 2019

WORK EXPERIENCE

- **Senior Staff Research Scientist** at Visa Inc. 2023 - Current
 - **Lead of the Digital Currency Team**, driving innovations in CBDC interoperability and secure payment systems
 - Directed pioneering research in interoperable and privacy-preserving payment protocols and smart contracts
- **Staff Research Scientist** at **Visa Inc.**, Security and Digital Currency Team 2020 - 2023
 - Developed scalable blockchain solutions, including programmable payment channels
- **Research Scientist Intern** at **Visa Inc.**, Blockchain Team 2019
- **Data Scientist Intern** at **Microsoft**, Cloud + AI, Xbox Knowledge Platform Team 2018
 - Built a structured user profile database and trained ML models to identify users likely to cancel Game Pass
- **Program Manager Intern** at **Microsoft**, Universal Store, Fraud Detection Team 2017
 - Integrated the Customer Knowledge Platform to enhance risk assessment with graph-based insights
- **Software Engineering Intern** at **Microsoft**, Universal Store, Fraud Detection Team 2016
 - Optimized fraud detection models, increased the profit efficiency by 2%, saving \$6M in fraud

SELECTED PROJECTS

- **Visa Tokenized Asset Platform** (Link)
Technologies: CBDC, Tokenized Deposits, Digital Settlement, Hyperledger Besu, Solidity
 - Led protocol development of Visa Tokenized Asset Platform
 - Oversaw BBVA and HKMAs e-HKD pilots to validate CBDC use cases and tokenized deposits (Link)
 - Implemented smart contracts for tokenization and intra-bank and inter-bank transfers
 - Collaborated with HSBC and Hang Seng Bank on digital settlement and the e-HKD sandbox
- **Blockchain-Based Programmable Finance Platform**
Technologies: Ethereum, Smart Contracts, NFT, CBDC, Solidity
 - Led protocol development of Visas prototype of Brazil's Real Digital CBDC for asset tokenization and auctioning
 - Implemented an on-chain auction and integrated Visas Universal Payments Channel for currency interoperability
 - Addressed challenges in cross-currency payments, asset tokenization, and smart contract automation
- **Programmable Payment Channels and Scalable Blockchain Auctions** (Link)
Technologies: Ethereum, Smart Contracts, Payment Channels, zk-SNARKs
 - Developed off-chain payment channels for secure, programmable transactions
 - Enabled dynamic enforcement of payment conditions without the need for on-chain verification
 - Designed a scalable auction system reducing blockchain transactions from n (# bidders) to only 3 in total
 - Ensured bid privacy and security using zk-SNARKs under adversarial conditions

SELECTED PATENTS

- **Fast Sync Blockchain System and Method** ([Link](#)) 2024
- **System, Method, and Computer Program Product for Preventing Miner Extractable Value (MEV) Attacks in a Blockchain Network** ([Link](#)) 2023
- **Privacy-Preserving Detection for Directional Electronic Communications** ([Link](#)) 2023
- **Universal Payment Channel System and Method** ([Link](#)) 2023
- **Conditional Offline Interaction System and Method** ([Link](#)) 2022
- **Accounting for Uncertainty When Calculating Profit Efficiency** ([Link](#)) 2019

SELECTED PUBLICATIONS

- **Exploring the Interplay Between Interaction Experience and Security Perception of Payment Authentication in Virtual Reality** ([Link](#)) 2024
IEEE Conference on Virtual Reality and 3D User Interfaces
- **Programmable Payment Channels** ([Link](#)) 2024
International Conference on Applied Cryptography and Network Security (ACNS)
- **A Plug-and-Play Long-Range Defense System for Proof-of-Stake Blockchains** ([Link](#)) 2024
European Symposium on Research in Computer Security (ESORICS)
- **Tokenization of Commodity Receivable** ([Link](#)) 2024
- **Uncovering Impact of Mental Models Towards Adoption of Multi-Device Crypto-Wallets** ([Link](#)) 2023
ACM Conference on Computer and Communications Security (CCS)
- **Unlinkability and Interoperability in Account-Based Universal Payment Channels** ([Link](#)) 2023
International Conference on Financial Cryptography and Data Security (FC)
- **How Interactions Influence Users' Security Perception of Virtual Reality Authentication?** ([Link](#)) 2023
- **SoK: Web3 Recovery Mechanisms** ([Link](#)) 2023
- **Scalable Off-Chain Auctions** ([Link](#)) 2023
- **Towards Overcoming the Undercutting Problem** ([Link](#)) 2022
International Conference on Financial Cryptography and Data Security (FC)
- **Empirical Understanding of Deletion Privacy: Experiences, Expectations, and Measures** ([Link](#)) 2022
USENIX Security Symposium
- **All Your Credentials Are Belong To Us: On Insecure WPA2-Enterprise Configurations** ([Link](#)) 2021
Conference on Computer and Communications Security (CCS)
- **Universal Payment Channels: An Interoperability Platform for Digital Currencies** ([Link](#)) 2021
- **Deceptive Deletions for Protection of Withdrawn Posts on Social Media** ([Link](#)) 2020
Network and Distributed System Security Symposium (NDSS)
- **Towards a Two-Tier Hierarchical Infrastructure: An Offline Payment System for Central Bank Digital Currencies** ([Link](#)) 2020
- **MoneyMorph: Censorship Resistant Rendezvous using Permissionless Cryptocurrencies** ([link](#)) 2020
Proceedings on Privacy Enhancing Technologies (PoPETs)
- **A Tale of Two Trees: One Writes, and Other Reads** ([link](#)) 2020
Proceedings on Privacy Enhancing Technologies (PoPETS)

TECHNICAL SKILLS

- **Blockchain & Cryptocurrencies:** CBDC Architecture, Permissioned Networks, Tokenization, NFTs, Decentralized Exchanges, Ethereum, Bitcoin, Smart Contracts, Payment Channels, State Channels, Rollups
- **Security & Privacy:** Fraud Detection, Privacy-Preserving Systems, Cryptographic Protocols, zk-SNARKs
- **Programming:** Solidity, Python, Java, C
- **Tools & Frameworks:** Foundry, Brownie, Ganache, Truffle, Neo4j